

13 SPI - Serial Peripheral Interface

13.1 Introduction

Main functions

The serial peripheral interface bus or SPI bus is a synchronous serial data link that operates in full duplex mode. The TLF35584 communicates in slave mode where the master (μ C) initiates the data frame. The TLF35584 should be addressed via a dedicated chip select line. This allows a connection of other slave devices to the SPI bus.

Data transmission

To begin a communication, the μ C first configures the clock, using a frequency less than or equal to the maximum frequency the TLF35584 supports. The μ C pulls down the chip select for the TLF35584.

Functional description

SPI basic access: All data on MOSI (pin SDI) is captured on the rising edge of SPI clock signal (pin SCL) and shifted on the falling edge of SPI clock signal (pin SCL). The same methodology needs to be applied in the SPI master for MISO (SDO). A read operation has to start with CMD-bit being 1'b0 and a write operation has to start with CMD-bit being 1'b1.

In case of a write operation is performed the written command to SDI is looped back to SDO.

The parity is calculated for the output data stream in case a read operation is performed. The data for the calculation consists of 1'b1, status [5:0] and rd_data[7:0]. The parity bit is set to '1' if the number of '1' in the output data stream is odd, i.e. XOR function between all 15 bits to send out.

Parity is checked on write data. The parity is calculated on the incoming bit stream for the cmd bit, the six address bits and the eight data bits.

Configuration via SPI can be done anytime, if the state machine (FSM) is in INIT state, NORMAL state, WAKE state or SLEEP state. During SLEEP state the SPI has a decreased maximum clock frequency, for details please refer to [Table 19](#).

SPI access timing can be found in the following diagram:

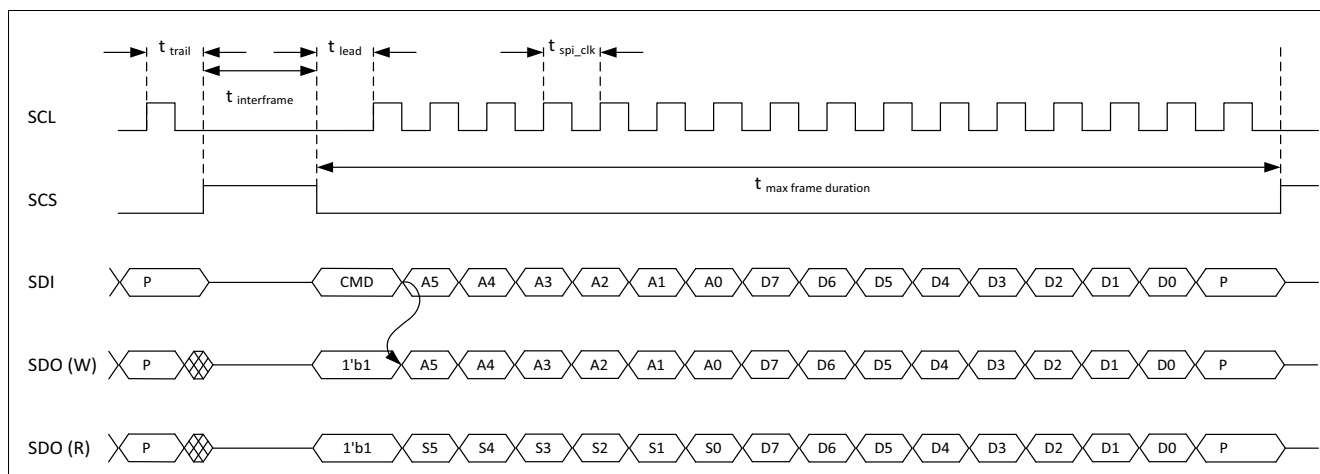


Figure 74 SPI - frame in normal mode

SPI MISO:

- During write, data from MOSI is directly looped back, during read, the addressed register content will be provided in the very same SPI frame
- The cmd bit is always set to 1'b1. All other status bits are set to zero.

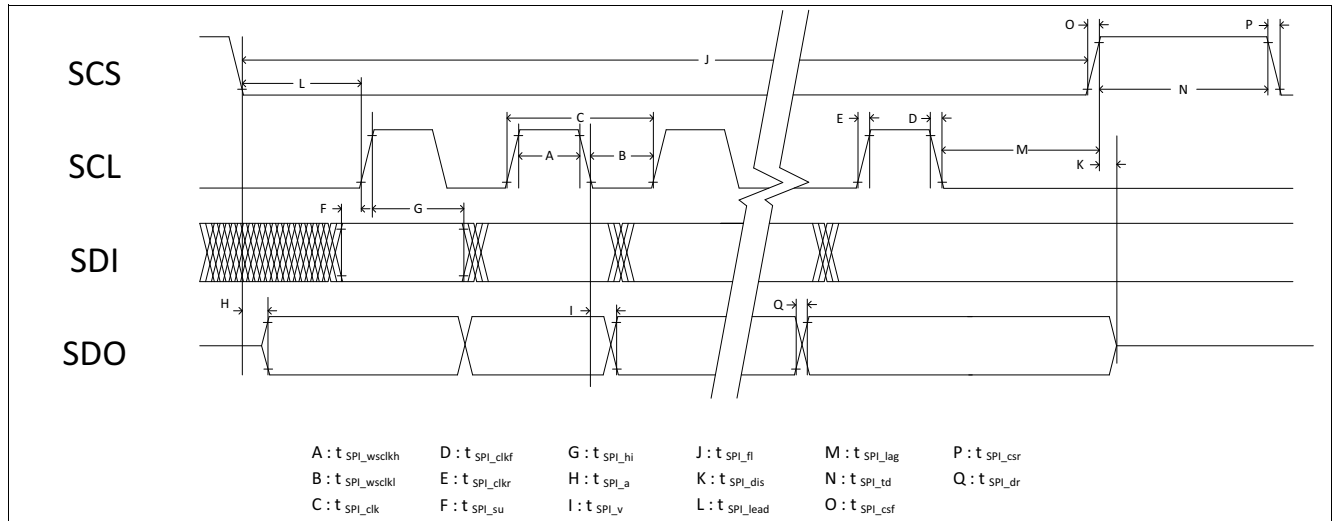


Figure 75 SPI - Timings

Table 19 Electrical Characteristics: SPI - Timings

$V_{VS} = 6.0 \text{ V to } 40 \text{ V}$, $T_j = -40^\circ\text{C to } +150^\circ\text{C}$, all voltages with respect to ground, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note / Test Condition	Number
		Min.	Typ.	Max.			
CLK_SPI Operating Frequency	f_{SPI_clk}	—	—	10	MHz	1)	P_13.1.1
CLK_SPI Operating Frequency	f_{SPI_clk}	—	—	1.5	MHz	SLEEP state	P_13.1.2
CLK Signal duty cycle	D_{SCL}	45	50	55	%		P_13.1.3
CLK_SPI Operating Period	t_{SPI_clk}	100	—	—	ns		P_13.1.4
CLK_SPI High Time	t_{SPI_wscikh}	45	—	—	ns		P_13.1.5
CLK_SPI Low Time	t_{SPI_wscikl}	45	—	—	ns		P_13.1.6
CLK_SPI Fall Time	t_{SPI_clkh}	—	—	$0.1 * t_{SPI_fact}$	[ns]	$t_{SPI_fact} \leq t_{SPI_clk}$; $100 \text{ ns} \leq t_{SPI_fact} \leq 1 \mu\text{s}$	P_13.1.7
CLK_SPI Rise Time	t_{SPI_clkr}	—	—	$0.1 * t_{SPI_fact}$	[ns]	$t_{SPI_fact} \leq t_{SPI_clk}$; $100 \text{ ns} \leq t_{SPI_fact} \leq 1 \mu\text{s}$	P_13.1.8
CLK_SPI Lead Time	t_{SPI_lead}	100	—	—	ns		P_13.1.9
CLK_SPI Lag Time	t_{SPI_lag}	50	—	—	ns		P_13.1.10
SPI Chip Select (SCS) Rise Time	t_{SPI_csr}	—	—	$0.2 * t_{SPI_fact}$	[ns]	$t_{SPI_fact} \leq t_{SPI_lag}$; $50 \text{ ns} \leq t_{SPI_fact} \leq 500 \text{ ns}$	P_13.1.11

Table 19 Electrical Characteristics: SPI - Timings (cont'd)

$V_{VS} = 6.0 \text{ V to } 40 \text{ V}$, $T_j = -40^\circ\text{C to } +150^\circ\text{C}$, all voltages with respect to ground, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note / Test Condition	Number
		Min.	Typ.	Max.			
SPI Chip Select (SCS) Fall Time	$t_{\text{SPI_csf}}$	—	—	$0.2 \cdot t_{\text{SPI_fact}}$	[ns]	$t_{\text{SPI_fact}} \leq t_{\text{SPI_lead}}$; $100 \text{ ns} \leq t_{\text{SPI_fact}} \leq 1 \mu\text{s}$	P_13.1.12
SPI Data Input (SDI) Setup	$t_{\text{SPI_su}}$	10	—	—	ns		P_13.1.13
SPI Data Input (SDI) Hold Time	$t_{\text{SPI_hi}}$	10	—	—	ns		P_13.1.14
SPI Data Output (SDO) Valid after CLK_SPI	$t_{\text{SPI_v}}$	—	—	$(0.1 \cdot t_{\text{SPI_fact}}) + 36$	[ns]	$C_{\text{SDO,load}} = 50 \text{ pF}$; $t_{\text{SPI_fact}} \geq t_{\text{SPI_clkf}}$; $10 \text{ ns} \leq t_{\text{SPI_fact}} \leq 100 \text{ ns}$	P_13.1.15
SPI Write Propagation Delay SDI to SDO	$t_{\text{SPI_wpd}}$	—	—	35	ns		P_13.1.16
SPI Data Output (SDO) Access	$t_{\text{SPI_a}}$	—	—	50	ns	$C_{\text{SDO,load}} = 50 \text{ pF}$	P_13.1.17
SPI Data Output (SDO) Disable Time	$t_{\text{SPI_dis}}$		—	100	ns	$C_{\text{SDO,load}} = 50 \text{ pF}$	P_13.1.18
Sequential Transfer Delay	$t_{\text{SPI_td}}$	350	—	—	ns		P_13.1.19
Sequential Transfer Delay	$t_{\text{SPI_td}}$	2	—	—	μs	SLEEP state	P_13.1.20
Frame duration (SCS low)	$t_{\text{SPI_fl}}$		—	1.85	ms		P_13.1.21

1) For max. achievable CLK_SPI operating frequency, please consider the CLK_SPI rise- and fall times ($t_{\text{SPI_clkr}}$ and $t_{\text{SPI_clkf}}$).

SPI errors:

- Wrong parity bit during write, write data is ignored.
- Write to invalid address, write data is ignored.
- Wrong number of SPI clock cycles while SCS is low, write data is ignored, read data is provided by the device with each SPI clock cycle.
- Read from invalid address (all data bits zero returned on MISO for read data). For this case the parity bit is inverted/corrupted after complete calculation by the device.
- Invalid frame duration, write data is ignored and the output driver for SDO is turned off internally by the device after $t_{\text{SPI_fl}}$. Read data is provided with each SPI clock cycle as long as SCS is low for less than $t_{\text{SPI_fl}}$.
- If the number of SPI clock cycles is different than 16 and an invalid frame duration error is detected by the device, the invalid frame duration status flag is set and the wrong number of SPI clock cycle status flag is set.

In case of an SPI error occurs, an interrupt will be generated.

Interrupts on SPI errors are initiated only after SCS has been driven high or the frame time-out occurred.

13.2 SPI Write Access To Protected Registers

Certain internal registers (**SYSPCFG0**, **SYSPCFG1**, **WDCFG0**, **WDCFG1**, **FWDCFG**, **WWDCFG0**, **WWDCFG1**) need to be protected against being overwritten accidentally. The status of the protection can be checked by reading the **LOCK** bit in the register **PROTSTAT**.

Write access to these registers is only possible after a dedicated 32 bit UNLOCK sequence has been sent via SPI. The four bytes need to be send without any other SPI write access in between. Error in doing so will reset the sequence detection, i.e. a new UNLOCK sequence has to be send. An interrupt is generated and the number of successfully detected UNLOCK sequence bytes is set to zero if a write access to any other register then **PROTCFG** is detected in between. The access to the protected registers is possible in INIT, NORMAL and WAKE state.

The UNLOCK sequence consists of a 32-bit sequence of 4 consecutive bytes (1: 0xAB; 2:0xEF; 3:0x56; 4:0x12) which have to be sent with no other SPI write access in between. The correctness of every written byte can be checked by reading the register **PROTSTAT**. Once the UNLOCK sequence has been performed successfully, any protected configuration request register can be written. In order to ensure proper writing to the protected configuration request registers the microcontroller shall read back the register values and verify the correctness by checking data. The data bits written to the protected configuration request registers are send back inverted during read operation, that means the microcontroller can calculate an XOR of the register data read and expected. The results should be 0xFF in case of correct register data. The TLF35584 will not check the correctness of the values in the register.

All protected configuration request register values are captured by the respective functions only after a successful LOCK sequence has been performed. A successful LOCK sequence consists of a 32-bit sequence of 4 consecutive bytes (1: 0xDF; 2:0x34; 3:0xBE; 4:0xCA) which have to be send with no other SPI write access in between. The correctness of every written byte can be checked by reading the register **PROTSTAT**.

Error in doing so will reset the sequence detection, i.e. a new LOCK sequence has to be sent. In this case (any SPI write access in between LOCK sequence) an interrupt is generated.

Upon detection of a successful LOCK sequence the configuration registers and all internal functions are updated with the values from the protected configuration request registers. It is the responsibility of the uC to ensure all registers are configured properly by either writing a new value into a particular register or by reading back a register which is supposed to be unchanged. Partial reconfiguration of the protected registers, i.e. configuring just a single function and leaving other functions unchanged is not supported as with the successful LOCK sequence all protected configuration request registers are taken over into the configuration (**RSYSPCFG0**, **RSYSPCFG1**, **RWDCFG0**, **RWDCFG1**, **RFWDCFG**, **RWWDCFG0**, **RWWDCFG1**).

After the LOCK sequence an internal configuration time of max. 60 μ s has to be considered to ensure that the new configuration is taken over.

Affected functions:

- All watchdog configuration registers for WWD and FWD.
- Enabling of WD while the device is in SLEEP state
- All Error pin monitoring configuration registers
- Enabling of Error pin monitoring while the FSM is in SLEEP
- A dedicated register to enable or disable the STDBY LDO
- Configuration of SS2 delay time failure events not leading to FAILSAFE state

Read access to any protected configuration request register is always possible.

13.3 SPI Write Initiated State Transition Request And Regulator Configuration

A State machine transition can be initiated via SPI command(s). In case the state of any selectable voltage source (post regular) is expected to change for the next state, this information has to be sent together with the command into the same register. In case the setting for a particular voltage source (post regulator) is supposed to change but the state needs to be unchanged, the same approach can be applied. This basically means, the SPI command contains the current state of the FSM but a different setting for configurable voltage source (post regulator).

In order to request a state transition and/or a change of the LDO configuration the request data have to be written to two separated registers **DEVCTRL** and **DEVCTRLN** after each other consequently. The data written to DEVCTRLN have to be inverted bitwise compared to the data written to DEVCTRL. The request will be only accepted when the two registers are written consecutively after each other (first **DEVCTRL** and second **DEVCTRLN**) and will be taken over with the rising edge of the CS at the end of the second command.

In case of an invalid request (wrong sequence or DEVCTRLN not inverted to DEVCTRL) it will be rejected, an interrupt is generated and the corresponding status flag (**NO_OP**) is set. Incase of an invalid state transition request according to the **State Machine** in **Chapter 11** the request is ignored without issuing an interrupt.

13.4 Registers Description

Table 20 Abbreviations

*R0)	Registers that are being reset only in case of a POR.
*R1)	Registers that are being reset only in case of STANDBY and a POR.
*R2)	Registers that are being reset only in case of FAILSAFE, STANDBY and a POR.
*R3)	Registers that are being reset in case of “Move to INIT” event, FAILSAFE, STANDBY and a POR.
r	Bits that are readable (read)
rw	Bits that are readable and writable (read-write)
rwp	Bits that are readable and writable but protected by register PROTCFG (read-write-protected)
rw1c	Bits that are readable and to clear the bit you have to write a 1 to it. (read-write-1-to-clear). Flag-bits are updated based on the occurred condition.
rwhc	Bits that are readable and writable after writing the operation is triggered, once this is done successfully the bit is cleared by hardware. (read-write-hardware-cleared)
rwhu	Bits that are readable and writable, after the operation the bit is updated by hardware. (read-write-hardware-updated)

Table 21 Register Address Space

Module	Base Address	End Address	Note
BusInterface	0 _H	3F _H	Slave interface

Table 22 Register Overview

Register Short Name	Register Long Name	Offset Address	Reset Value
DEVCFG0	Device configuration 0 *R2)	00 _H	08 _H
DEVCFG1	Device configuration 1 *R0)	01 _H	06 _H
DEVCFG2	Device configuration 2 *R2)	02 _H	00 _H
PROTCFG	Protection register *R2)	03 _H	00 _H
SYSPCFG0	Protected System configuration request 0 *R1)	04 _H	01 _H
SYSPCFG1	Protected System configuration request 1 *R2)	05 _H	00 _H
WDCFG0	Protected Watchdog configuration request 0 *R2)	06 _H	9B _H
WDCFG1	Protected Watchdog configuration request 1 *R2)	07 _H	09 _H
FWDCFG	Protected Functional watchdog configuration request *R2)	08 _H	0B _H
WWDCFG0	Protected Window watchdog configuration request 0 *R2)	09 _H	06 _H
WWDCFG1	Protected Window watchdog configuration request 1 *R2)	0A _H	0B _H
RSYSPCFG0	System configuration 0 status *R0)	0B _H	01 _H
RSYSPCFG1	System configuration 1 status *R3) ¹⁾	0C _H	00 _H
RWDCFG0	Watchdog configuration 0 status *R3)	0D _H	9B _H

Table 22 Register Overview (cont'd)

Register Short Name	Register Long Name	Offset Address	Reset Value
RWDCFG1	Watchdog configuration 1 status *R3)	0E _H	09 _H
RFWDCFG	Functional watchdog configuration status *R3)	0F _H	0B _H
RWWDCFG0	Window watchdog configuration 0 status *R3)	10 _H	06 _H
RWWDCFG1	Window watchdog configuration 1 status *R3)	11 _H	09 _H
WKTIMCFG0	Wake timer configuration 0 *R2)	12 _H	00 _H
WKTIMCFG1	Wake timer configuration 1 *R2)	13 _H	00 _H
WKTIMCFG2	Wake timer configuration 2 *R2)	14 _H	00 _H
DEVCTRL	Device control request *R2)	15 _H	00 _H
DEVCTRLN	Device control inverted request *R2)	16 _H	00 _H
WWDSCMD	Window watchdog service command *R2)	17 _H	00 _H
FWDRSP	Functional watchdog response command *R2)	18 _H	00 _H
FWDRSPSYNC	Functional watchdog response command with synchronization *R2)	19 _H	00 _H
SYSFAIL	Failure status flags *R1)	1A _H	00 _H
INITERR	Init error status flags *R2)	1B _H	00 _H
IF	Interrupt flags *R2)	1C _H	00 _H
SYSSF	System status flags *R2)	1D _H	00 _H
WKSF	Wakeup status flags *R2)	1E _H	00 _H
SPISF	SPI status flags *R2)	1F _H	00 _H
MONSF0	Monitor status flags 0 *R1)	20 _H	00 _H
MONSF1	Monitor status flags 1 *R1)	21 _H	00 _H
MONSF2	Monitor status flags 2 *R2)	22 _H	00 _H
MONSF3	Monitor status flags 3 *R1)	23 _H	00 _H
OTFAIL	Over temperature failure status flags *R1)	24 _H	00 _H
OTWRNSF	Over temperature warning status flags *R2)	25 _H	00 _H
VMONSTAT	Voltage monitor status *R2)	26 _H	00 _H
DEVSTAT	Device status *R2)	27 _H	00 _H
PROTSTAT	Protection status *R1)	28 _H	01 _H
WWDSTAT	Window watchdog status *R3)	29 _H	00 _H
FWDSTAT0	Functional watchdog status 0 *R3)	2A _H	30 _H
FWDSTAT1	Functional watchdog status 1 *R3)	2B _H	00 _H
ABIST_CTRL0	ABIST control0 *R2) ¹⁾	2C _H	00 _H
ABIST_CTRL1	ABIST control1 *R2)	2D _H	00 _H
ABIST_SELECT0	ABIST select 0 *R2)	2E _H	00 _H
ABIST_SELECT1	ABIST select 1 *R2)	2F _H	00 _H
ABIST_SELECT2	ABIST select 2 *R2)	30 _H	00 _H
GTM	Global testmode *R2)	3F _H	02 _H
BCK_FREQ_CHANGE	Buck switching frequency change *R2)	31 _H	00 _H

Table 22 Register Overview (cont'd)

Register Short Name	Register Long Name	Offset Address	Reset Value
BCK_FRE_SPREAD	Buck Frequency spread *R2)	32 _H	00 _H
BCK_MAIN_CTRL	Buck main control *R2)	33 _H	00 _H

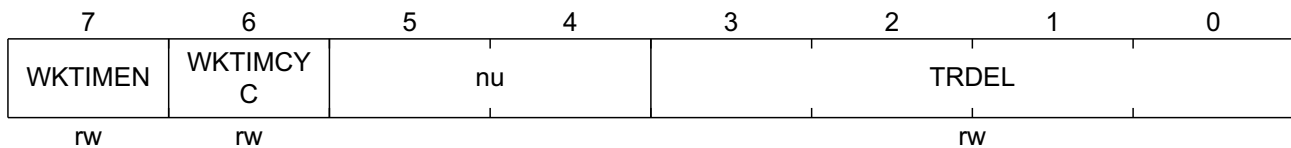
1) Reset class of single bits is different. See register for details.

The registers are addressed wordwise.

13.4.1 Device registers

Device configuration 0 *R2)

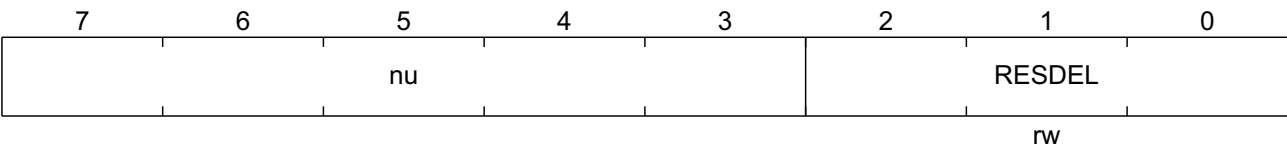
DEVCFG0	Offset	Reset Value
Device configuration 0 *R2)	00 _H	08 _H



Field	Bits	Type	Description
WKTIMEN	7	rw	Wake timer enable 0 _D Wake timer disabled 1 _D Wake timer enabled in SLEEP or STANDBY state Reset: 0 _H
WKTIMCYC	6	rw	Wake timer cycle period 0 _D 10 us 1 _D 10 ms Reset: 0 _H
nu	5:4	none	Reset: 0 _H
TRDEL	3:0	rw	Transition delay into low power states For STANDBY and SLEEP transition. Defined as a step of 100 us. 0 _D 100 us 1 _D 200 us 2 _D 300 us ... 15 _D 1600 us Reset: 8 _H

Device configuration 1 *R0)

DEVCFG1	Offset	Reset Value
Device configuration 1 *R0)	01 _H	06 _H



Field	Bits	Type	Description
nu	7:3	none	Reset: 00 _H
RESDEL	2:0	rw	Reset release delay time 0 _D 200 us 1 _D 400 us 2 _D 800 us 3 _D 1ms 4 _D 2 ms 5 _D 4 ms 6 _D 10 ms 7 _D 15 ms Reset: 6 _H

Device configuration 2 *R2)

DEVCFG2

Offset

Reset Value

Device configuration 2 *R2)

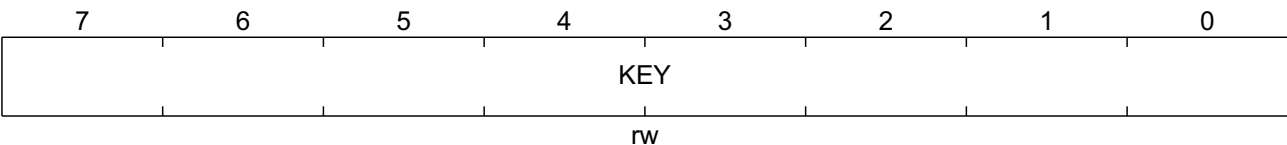
02_H00_H

7	6	5	4	3	2	1	0
EVCEN	STU	FRE	CMONEN	CTHR		ESYNPHA	ESYNEN
r	r	r	rw	rw		rw	rw

Field	Bits	Type	Description
EVCEN	7	r	External core supply enable status 0 _D External core supply disabled 1 _D External core supply enabled Reset: 0 _H
STU	6	r	Step-up converter enable status 0 _D Disabled 1 _D Enabled Reset: 0 _H
FRE	5	r	Step-down converter frequency selection status 0 _D Step-down converter runs on low frequency range 1 _D Step-down converter runs on high frequency range Reset: 0 _H
CMONEN	4	rw	QUC current monitor enable for transition to a low power state For STANDBY and SLEEP transition. The setting is overwritten in SLEEP as current monitoring is always enabled. 0 _D Disabled 1 _D Enabled Reset: 0 _H
CTHR	3:2	rw	QUC current monitoring threshold value 0 _D 10 mA 1 _D 30 mA 2 _D 60 mA 3 _D 100 mA Reset: 0 _H
ESYNPHA	1	rw	External synchronization output phase 0 _D No phase shift 1 _D 180 phase shift Reset: 0 _H
ESYNEN	0	rw	Synchronization output for external switchmode regulator enable 0 _D Disable 1 _D Enable Reset: 0 _H

Protection register *R2)

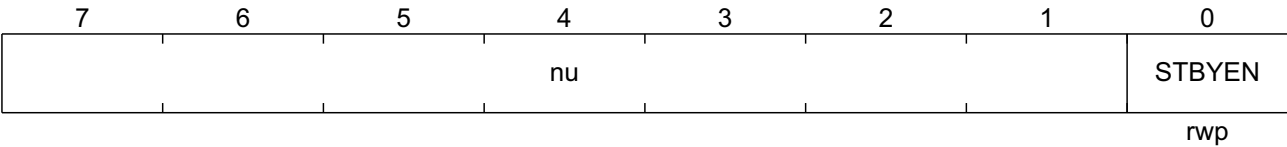
PROTCFG	Offset	Reset Value
Protection register *R2)	03 _H	00 _H



Field	Bits	Type	Description
KEY	7:0	rw	Protection key Protection key register to request write access to protected registers. Unlock: write 32-bit sequence of 4 consecutive bytes (1: 0xAB 2:0xEF 3:0x56 4:0x12) to unlock access to protected registers. Lock: write 32-bit sequence of 4 consecutive bytes (1: 0xDF 2:0x34 3:0xBE 4:0xCA) to lock access to protected registers. All configured values are applied to SSC and WD module after the lock. AB _H Key 1 to unlock protected registers. EF _H Key 2 to unlock protected registers. 56 _H Key 3 to unlock protected registers. 12 _H Key 4 to unlock protected registers. DF _H Key 1 to lock protected registers. 34 _H Key 2 to lock protected registers. BE _H Key 3 to lock protected registers. CA _H Key 4 to lock protected registers. Reset: 00 _H

Protected System configuration request 0 *R1)

SYSPCFG0	Offset	Reset Value
Protected System configuration request 0 *R1)	04 _H	01 _H



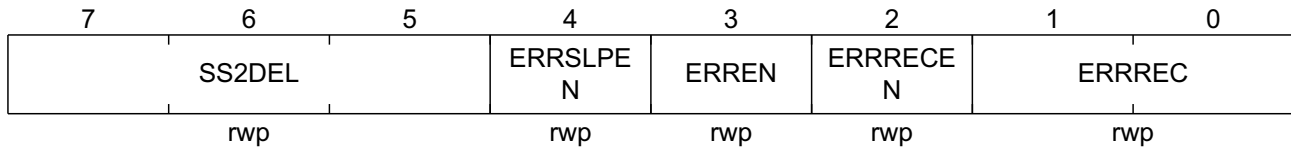
Field	Bits	Type	Description
nu	7:1	none	not used bits shall be written as 0 and will always return 1 upon read Reset: 00 _H
STBYEN	0	rwp	Request standby regulator QST enable Valid for all device states except FAILSAFE. 0 _D Disabled 1 _D Enabled Reset: 1 _H

Protected System configuration request 1 *R2)

SYSPCFG1

Offset

Reset Value

Protected System configuration request 1
*R2)05_H08_H

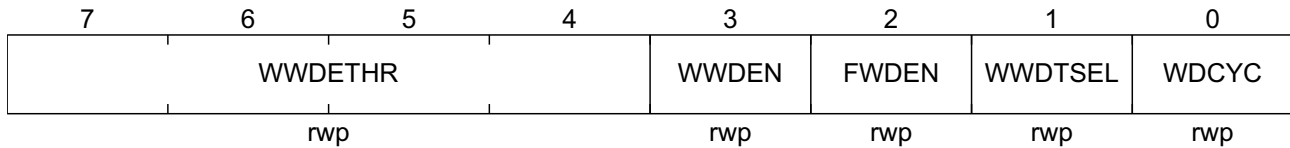
Field	Bits	Type	Description
SS2DEL	7:5	rwp	Request safe state 2 delay Applied for transitions from NORMAL to INIT, WAKE and SLEEP state. 0 _D no delay 1 _D 10 ms 2 _D 50 ms 3 _D 100 ms 4 _D 250 ms Reset: 0 _H
ERRSLPEN	4	rwp	Request ERR pin monitor functionality enable while the system is in SLEEP 0 _D ERR pin monitor is disabled in SLEEP 1 _D ERR pin monitor can be active in SLEEP depending on ERREN bit value. Reset: 0 _H
ERREN	3	rwp	Request ERR pin monitor enable 0 _D Disabled 1 _D Enabled Reset: 1 _H
ERRRECEN	2	rwp	Request ERR pin monitor recovery enable 0 _D Disabled 1 _D Enabled Reset: 0 _H
ERRREC	1:0	rwp	Request ERR pin monitor recovery time 0 _D 1 ms 1 _D 2.5 ms 2 _D 5 ms 3 _D 10 ms Reset: 0 _H

Protected Watchdog configuration request 0 *R2)

WDCFG0

Offset

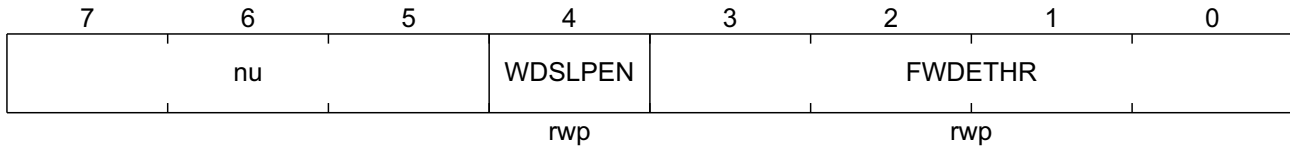
Reset Value

Protected Watchdog configuration request 0
*R2)06_H9B_H

Field	Bits	Type	Description
WWDETHR	7:4	rwp	Request window watchdog error threshold WWD error threshold to generate reset and enter into INIT state. Reset: 9 _H
WWDEN	3	rwp	Request window watchdog enable 0 _D Disabled 1 _D Enabled Reset: 1 _H
FWDEN	2	rwp	Request functional watchdog enable 0 _D Disabled 1 _D Enabled Reset: 0 _H
WWDTSSEL	1	rwp	Request window watchdog trigger selection This is ignored when window watchdog is disabled. 0 _D External WDI input used as a WWD trigger 1 _D WWD is triggered by SPI write to WWDSCMD register Reset: 1 _H
WDCYC	0	rwp	Request watchdog cycle time 0 _D 0,1 ms tick period 1 _D 1 ms tick period Reset: 1 _H

Protected Watchdog configuration request 1 *R2)

WDCFG1	Offset	Reset Value
Protected Watchdog configuration request 1 *R2)	07 _H	09 _H



Field	Bits	Type	Description
nu	7:5	none	not used bits shall be written as 0 and will always return 1 upon read Reset: 0 _H
WDSLPE	4	rwp	Request watchdog functionality enable while the device is in SLEEP 0 _D Disabled 1 _D Enabled, the WD will work based on individual configuration (WWDEN & FWDEN) settings while the system is in SLEEP mode Reset: 0 _H
FWDETHR	3:0	rwp	Request functional watchdog error threshold FWD error threshold to generate reset and enter into INIT state. Reset: 9 _H

Protected Functional watchdog configuration request *R2)

FWDCFG	Offset	Reset Value
Protected Functional watchdog configuration request *R2)	08 _H	0B _H

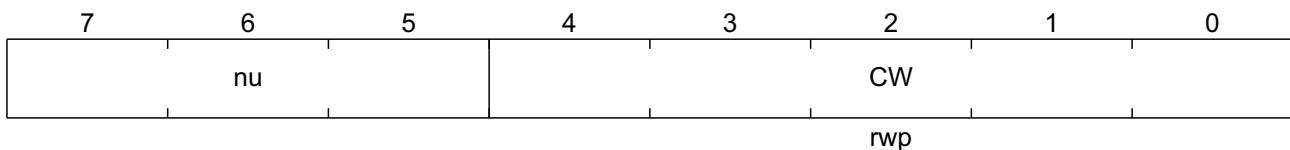


Field	Bits	Type	Description
nu	7:5	none	not used bits shall be written as 0 and will always return 1 upon read Reset: 0 _H

Field	Bits	Type	Description
WDHBTP	4:0	rwp	Request functional watchdog heartbeat timer period Defined as a multiple of 50 watchdog cycles (RWDCFG0.WDCYC). 0 _D 50 wd cycles 1 _D 100 wd cycles 2 _D 150 wd cycles ... 31 _D 1600 wd cycles Reset: 0B _H

Protected Window watchdog configuration request 0 *R2)

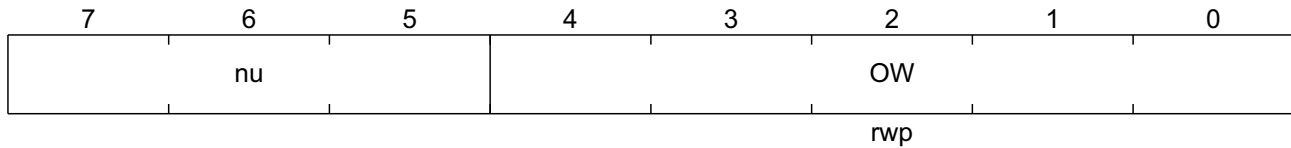
WWDCFG0	Offset	Reset Value
Protected Window watchdog configuration request 0 *R2)	09 _H	06 _H



Field	Bits	Type	Description
nu	7:5	none	not used bits shall be written as 0 and will always return 1 upon read Reset: 0 _H
CW	4:0	rwp	Request window watchdog closed window time Defined as a multiple of 50 watchdog cycles (RWDCFG0.WDCYC). 0 _D 50 wd cycles 1 _D 100 wd cycles 2 _D 150 wd cycles ... 31 _D 1600 wd cycles Reset: 06 _H

Protected Window watchdog configuration request 1 *R2)

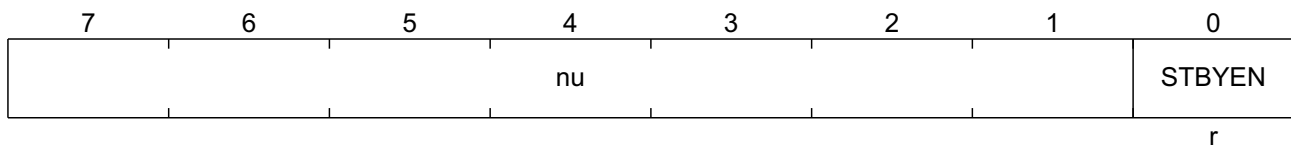
WWDCFG1	Offset	Reset Value
Protected Window watchdog configuration request 1 *R2)	0A_H	0B_H



Field	Bits	Type	Description
nu	7:5	none	not used bits shall be written as 0 and will always return 1 upon read Reset: 0 _H
OW	4:0	rwp	Request window watchdog open window time Defined as a multiple of 50 watchdog cycles (RWDCFG0.WDCYC). 0 _D 50 wd cycles 1 _D 100 wd cycles 2 _D 150 wd cycles ... 31 _D 1600 wd cycles Reset: 0B _H

System configuration 0 status *R0)

RSYSPCFG0	Offset	Reset Value
System configuration 0 status *R0)	0B_H	01_H



Field	Bits	Type	Description
nu	7:1	none	Reset: 00 _H
STBYEN	0	r	Standby regulator QST enable status Current configuration of standby regulator QST enable. Valid for all device states except FAILSAFE. 0 _D Disabled 1 _D Enabled Reset: 1 _H

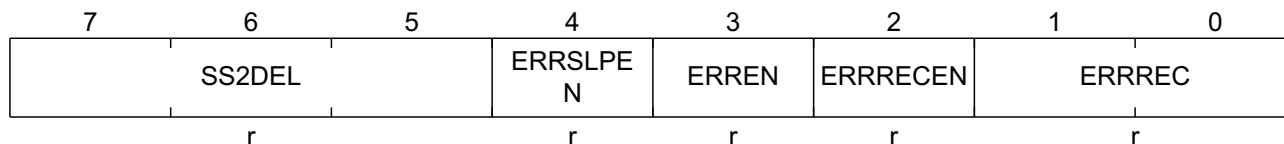
System configuration 1 status *R3) ¹⁾

RSYSPCFG1

Offset

Reset Value

System configuration 1 status *R3)

0C_H08_H

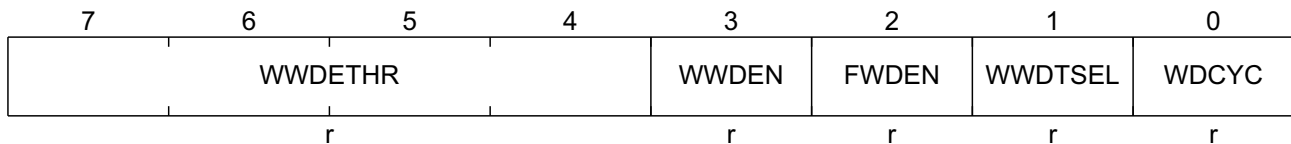
Field	Bits	Type	Description
SS2DEL	7:5	r	Safe state 2 delay status Current configuration of safe state 2 delay applied for transitions from NORMAL to INIT, WAKE and SLEEP state. Bits have different reset class than whole register. The Bits are reset according to *R1) 0 _D no delay 1 _D 10 ms 2 _D 50 ms 3 _D 100 ms 4 _D 250 ms Reset: 0 _H
ERRSLPEN	4	r	ERR pin monitor functionality enable status while the device is in SLEEP Current configuration of ERR pin monitor functionality enable for SLEEP. 0 _D ERR pin monitor is disabled in SLEEP 1 _D ERR pin monitor can be active in SLEEP depending on ERREN bit value. Reset: 0 _H
ERREN	3	r	ERR pin monitor enable status Current configuration of ERR pin monitor enable. 0 _D Disabled 1 _D Enabled Reset: 1 _H
ERRRECEN	2	r	ERR pin monitor recovery enable status Current configuration of ERR pin monitor recovery enable. 0 _D Disabled 1 _D Enabled Reset: 0 _H

1) Reset class of single bits is different. See register for details.

Field	Bits	Type	Description
ERRREC	1:0	r	ERR pin monitor recovery time status Current configuration of ERR pin monitor recovery time. 0 _D 1 ms 1 _D 2.5 ms 2 _D 5 ms 3 _D 10 ms Reset: 0 _H

Watchdog configuration 0 status *R3)

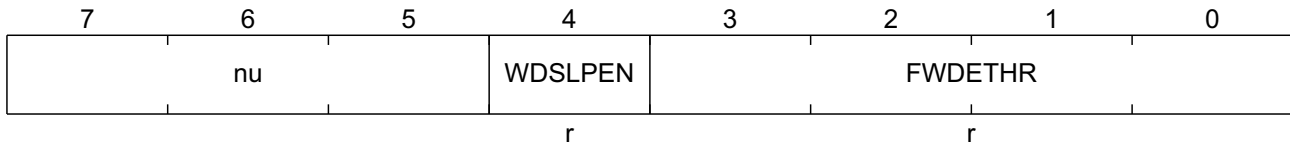
RWDCFG0	Offset	Reset Value
Watchdog configuration 0 status *R3)	0D _H	9B _H



Field	Bits	Type	Description
WWDETHR	7:4	r	Window watchdog error threshold status Current configuration of WWD error threshold to generate reset and enter into INIT state. Reset: 9 _H
WWDEN	3	r	Window watchdog enable status Current configuration of WWD enable. 0 _D Disabled 1 _D Enabled Reset: 1 _H
FWDEN	2	r	Functional watchdog enable status Current configuration of FWD enable. 0 _D Disabled 1 _D Enabled Reset: 0 _H
WWDTSEL	1	r	Window watchdog trigger selection status Current configuration of WWD trigger selection. This is ignored when window watchdog is disabled. 0 _D External WDI input used as a WWD trigger 1 _D WWD is triggered by SPI write to WWDSCMD register Reset: 1 _H
WDCYC	0	r	Watchdog cycle time status Current configuration of watchdog cycle time. 0 _D 0,1 ms tick period 1 _D 1 ms tick period Reset: 1 _H

Watchdog configuration 1 status *R3)

RWDCFG1	Offset	Reset Value
Watchdog configuration 1 status *R3)	0E_H	09_H



Field	Bits	Type	Description
nu	7:5	none	Reset: 0 _H
WDSLPE	4	r	Watchdog functionality enable status while the device is in SLEEP Current configuration of WD functionality enable for SLEEP. 0 _D Disabled 1 _D Enabled, the WD will work based on individual configuration (WWDEN & FWDEN) settings while the system is in SLEEP mode Reset: 0 _H
FWDETHR	3:0	r	Functional watchdog error threshold status Current configuration of FWD error threshold to generate reset and enter into INIT state. Reset: 9 _H

Functional watchdog configuration status *R3)

RFWDCFG	Offset	Reset Value
Functional watchdog configuration status *R3)	0F_H	0B_H

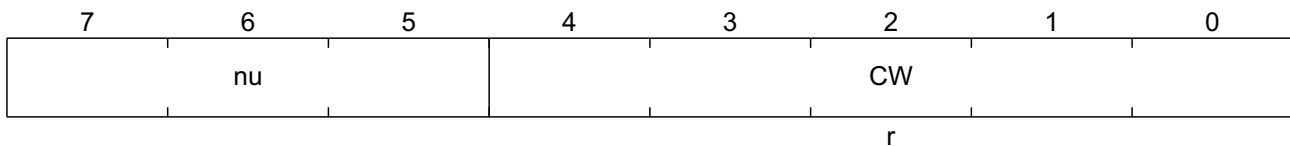


Field	Bits	Type	Description
nu	7:5	none	Reset: 0 _H

Field	Bits	Type	Description
WDHBTP	4:0	r	Functional watchdog heartbeat timer period status Current configuration of FWD heartbeat timer period defined as a multiple of 50 watchdog cycles (RWDCFG0.WDCYC). 0 _D 50 wd cycles 1 _D 100 wd cycles 2 _D 150 wd cycles .. _D ... 31 _D 1600 wd cycles Reset: 0B _H

Window watchdog configuration 0 status *R3)

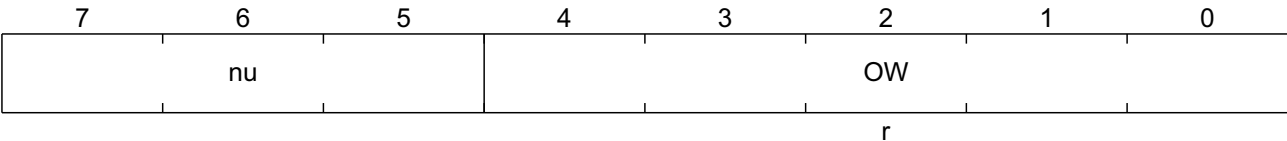
RWWDCFG0	Offset	Reset Value
Window watchdog configuration 0 status *R3)	10 _H	06 _H



Field	Bits	Type	Description
nu	7:5	none	Reset: 0 _H
CW	4:0	r	Window watchdog closed window time status Current configuration of WWD closed window time defined as a multiple of 50 watchdog cycles (RWDCFG0.WDCYC). 0 _D 50 wd cycles 1 _D 100 wd cycles 2 _D 150 wd cycles .. _D ... 31 _D 1600 wd cycles Reset: 06 _H

Window watchdog configuration 1 status *R3)

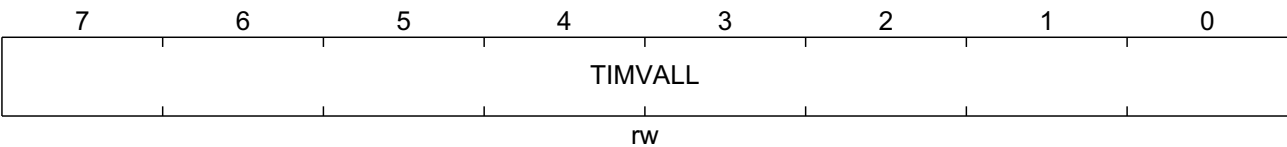
RWWDCFG1	Offset	Reset Value
Window watchdog configuration 1 status *R3)	11 _H	0B _H



Field	Bits	Type	Description
nu	7:5	none	Reset: 0 _H
OW	4:0	r	Window watchdog open window time status Current configuration of WWD open window time defined as a multiple of 50 watchdog cycles (RWDCFG0.WDCYC). 0 _D 50 wd cycles 1 _D 100 wd cycles 2 _D 150 wd cycles ... 31 _D 1600 wd cycles Reset: 0B _H

Wake timer configuration 0 *R2)

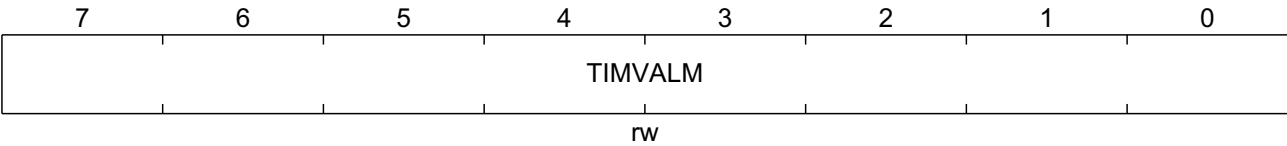
WKTIMCFG0	Offset	Reset Value
Wake timer configuration 0 *R2)	12 _H	00 _H



Field	Bits	Type	Description
TIMVALL	7:0	rw	Wake timer value lower bits Bits (7:0) of wake time defined as a multiple of wake timer cycles (DEVCFG0.WKTIMCYC). Reset: 00 _H

Wake timer configuration 1 *R2)

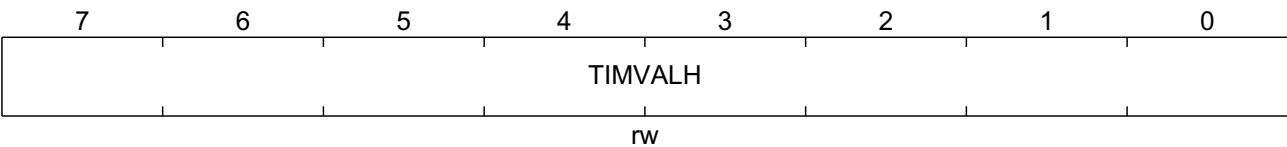
WKTIMCFG1	Offset	Reset Value
Wake timer configuration 1 *R2)	13 _H	00 _H



Field	Bits	Type	Description
TIMVALM	7:0	rw	Wake timer value middle bits Bits (15:8) of wake time defined as a multiple of wake timer cycles (DEVCFG0.WKTIMCYC). Reset: 00 _H

Wake timer configuration 2 *R2)

WKTIMCFG2	Offset	Reset Value
Wake timer configuration 2 *R2)	14 _H	00 _H



Field	Bits	Type	Description
TIMVALH	7:0	rw	Wake timer value higher bits Bits (23:16) of wake time defined as a multiple of wake timer cycles (DEVCFG0.WKTIMCYC). Reset: 00 _H

Device control request *R2)

DEVCTRL

Offset

Reset Value

Device control request *R2)

15_H00_H

7	6	5	4	3	2	1	0
TRK2EN	TRK1EN	COMEN	nu	VREFEN		STATEREQ	
rw	rw	rw		rw		rwhc	

Field	Bits	Type	Description
TRK2EN	7	rw	Request tracker2 QT2 enable 0 _D QT2 will be disabled after valid request 1 _D QT2 will be enabled after valid request Reset: 0 _H
TRK1EN	6	rw	Request tracker1 QT1 enable 0 _D QT1 will be disabled after valid request 1 _D QT1 will be enabled after valid request Reset: 0 _H
COMEN	5	rw	Request communication Ido QCO enable 0 _D QCO will be disabled after valid request 1 _D QCO will be enabled after valid request Reset: 0 _H
nu	4	none	Reset: 0 _H
VREFEN	3	rw	Request voltage reference QVR enable 0 _D QVR will be disabled after valid request 1 _D QVR will be enabled after valid request Reset: 0 _H
STATEREQ	2:0	rwhc	Request for device state transition Cleared to 000 by the HW after the request is processed. After writing a new state value a user should not change the value before it's cleared by HW. 7 _D RESERVED 6 _D RESERVED 5 _D WAKE 4 _D STANDBY 3 _D SLEEP 2 _D NORMAL 1 _D INIT 0 _D NONE Reset: 0 _H

Device control inverted request *R2)

DEVCTRLN

Offset

Reset Value

Device control inverted request *R2)

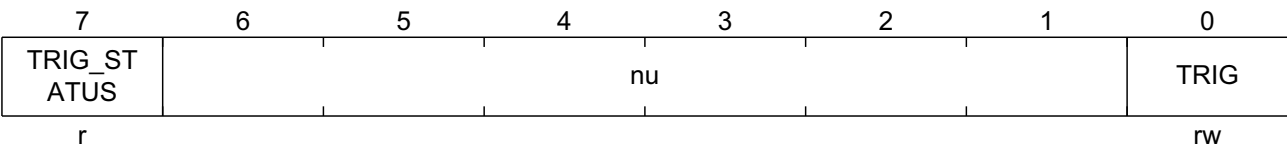
16_H00_H

7	6	5	4	3	2	1	0
TRK2EN	TRK1EN	COMEN	nu	VREFEN		STATEREQ	
rw	rw	rw		rw		rwhc	

Field	Bits	Type	Description
TRK2EN	7	rw	Request tracker2 QT2 enable 1 _D QT2 will be disabled after valid request 0 _D QT2 will be enabled after valid request Reset: 0 _H
TRK1EN	6	rw	Request tracker1 QT1 enable 1 _D QT1 will be disabled after valid request 0 _D QT1 will be enabled after valid request Reset: 0 _H
COMEN	5	rw	Request communication Ido QCO enable 1 _D QCO will be disabled after valid request 0 _D QCO will be enabled after valid request Reset: 0 _H
nu	4	none	Reset: 0 _H
VREFEN	3	rw	Request voltage reference QVR enable 1 _D QVR will be disabled after valid request 0 _D QVR will be enabled after valid request Reset: 0 _H
STATEREQ	2:0	rwhc	Request for device state transition Cleared to 000 by the HW after the request is processed. After writing a new state value a user should not change the value before it's cleared by HW. 7 _D NONE 6 _D INIT 5 _D NORMAL 4 _D SLEEP 3 _D STANDBY 2 _D WAKE 1 _D RESERVED 0 _D RESERVED Reset: 0 _H

Window watchdog service command *R2)

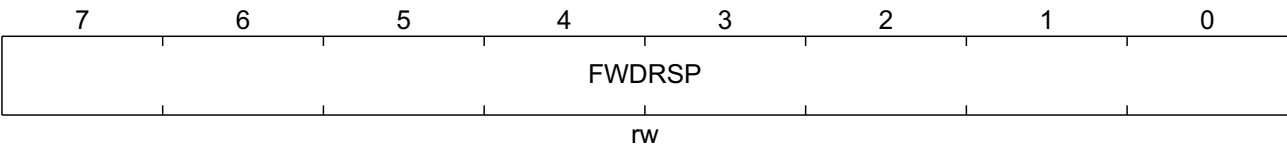
WWDSCMD	Offset	Reset Value
Window watchdog service command *R2)	17 _H	00 _H



Field	Bits	Type	Description
TRIG_STATUS	7	r	Last SPI trigger received Reset: 0 _H
nu	6:1	none	Reset: 00 _H
TRIG	0	rw	Window watchdog SPI trigger command Read TRIG_STATUS bit first and write inverted value to TRIG bit. Reset: 0 _H

Functional watchdog response command *R2)

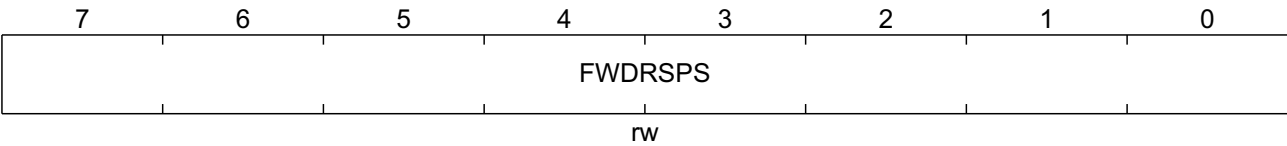
FWDRSP	Offset	Reset Value
Functional watchdog response command *R2)	18 _H	00 _H



Field	Bits	Type	Description
FWDRSP	7:0	rw	Functional watchdog response Write functional watchdog response bytes to this field. Reset: 00 _H

Functional watchdog response command with synchronization *R2)

FWDRSPSYNC	Offset	Reset Value
Functional watchdog response command with synchronization *R2)	19 _H	00 _H



Field	Bits	Type	Description
FWDRSPS	7:0	rw	Functional watchdog heartbeat synchronization response Write the last functional watchdog response byte to this field to synchronize/restart the heartbeat. Reset: 00 _H

Failure status flags *R1)

SYSFAIL

Failure status flags *R1)

Offset

1A_H

Reset Value

00_H

7	6	5	4	3	2	1	0
INITF	ABISTERR		nu		VMONF	OTF	VOLTSEL ERR
rw1c	rw1c				rw1c	rw1c	rw1c

Field	Bits	Type	Description
INITF	7	rw1c	INIT failure flag INIT failure due to the third INIT failure in row. i.e. The device restarts INIT phase from FAILSAFE. 0 _D No fault, write 0 - no action 1 _D Fault occurred, write 1 to clear the flags Reset: 0 _H
ABISTERR	6	rw1c	ABIST operation interrupted flag ABIST interrupted by any fault/event which is not part of ABIST. 0 _D No fault, write 0 - no action 1 _D Fault occurred, write 1 to clear the flags Reset: 0 _H
nu	5:3	none	Reset: 0 _H
VMONF	2	rw1c	Voltage monitor failure flag Voltage monitor failure occurred which lead to FAILSAFE. 0 _D No fault, write 0 - no action 1 _D Fault occurred, write 1 to clear the flags, read MONSF0 , MONSF1 and MONSF3 for details Reset: 0 _H
OTF	1	rw1c	Over temperature failure flag 0 _D No fault, write 0 - no action 1 _D Fault occurred, write 1 to clear the flags, read OTFAIL for details Reset: 0 _H
VOLTSELER R	0	rw1c	Double Bit error on voltage selection flag Device entered FAILSAFE state due to internal voltage selection failure. 0 _D No fault, write 0 - no action 1 _D Fault occurred, write 1 to clear the flags Reset: 0 _H

Init error status flags *R2)

INITERR

Init error status flags *R2)

Offset

1B_H

Reset Value

00_H

7	6	5	4	3	2	1	0
HARDRES	SOFTRES	ERRF	FWDF	WWDF	VMONF	nu	
rw1c	rw1c	rw1c	rw1c	rw1c	rw1c		

Field	Bits	Type	Description
HARDRES	7	rw1c	Hard reset flag Hard reset has been generated due to the second INIT failure in row. i.e. The device restarts INIT phase for the 3rd time. 0 _D No fault, write 0 - no action 1 _D Fault occurred, write 1 to clear the flags Reset: 0 _H
SOFTRES	6	rw1c	Soft reset flag Soft reset has been generated due to the first INIT failure. i.e. The device restarts INIT phase for the 2nd time. 0 _D No fault, write 0 - no action 1 _D Fault occurred, write 1 to clear the flags Reset: 0 _H
ERRF	5	rw1c	MCU error monitor failure flag 0 _D No fault, write 0 - no action 1 _D Fault occurred, write 1 to clear the flags Reset: 0 _H
FWDF	4	rw1c	Functional watchdog error counter overflow failure flag Functional watchdog error counter reached the error threshold. 0 _D No fault, write 0 - no action 1 _D Fault occurred, write 1 to clear the flags Reset: 0 _H
WWDF	3	rw1c	Window watchdog error counter overflow failure flag Window watchdog error counter reached the error threshold. 0 _D No fault, write 0 - no action 1 _D Fault occurred, write 1 to clear the flags Reset: 0 _H
VMONF	2	rw1c	Voltage monitor failure flag Voltage monitor failure occurred which lead to INIT. 0 _D No fault, write 0 - no action 1 _D Fault occurred, write 1 to clear the flags, read MONSF2 for details Reset: 0 _H
nu	1:0	rw1c	Reset: 0 _H

Interrupt flags *R2)

IF	Offset	Reset Value
Interrupt flags *R2)	1C _H	00 _H

7	6	5	4	3	2	1	0
INTMISS	ABIST	OTF	OTW	MON	SPI	WK	SYS
r	rw1c	rw1c	rw1c	rw1c	rw1c	rw1c	rw1c

Field	Bits	Type	Description
INTMISS	7	r	Interrupt not serviced in time flag Interrupt has not been serviced within $t_{INT,TO}$ time 0 _D No interrupt timeout happened 1 _D Interrupt timeout happened, cleared by hardware when all other flags in IF are cleared. Reset: 0 _H
ABIST	6	rw1c	Requested ABIST operation performed flag 0 _D No interrupt, write 0 - no action 1 _D Interrupt flag active, write 1 to clear the flag Reset: 0 _H
OTF	5	rw1c	Over temperature failure interrupt flag 0 _D No interrupt, write 0 - no action 1 _D Interrupt flag active, write 1 to clear the flag, read OTFAIL for details Reset: 0 _H
OTW	4	rw1c	Over temperature warning interrupt flag 0 _D No interrupt, write 0 - no action 1 _D Interrupt flag active, write 1 to clear the flag, read OTWRNSF for details Reset: 0 _H
MON	3	rw1c	Monitor interrupt flag 0 _D No interrupt, write 0 - no action 1 _D Interrupt flag active, write 1 to clear the flag, read MONSF0 , MONSF1 , MONSF2 and MONSF3 for details Reset: 0 _H
SPI	2	rw1c	SPI interrupt flag 0 _D No interrupt, write 0 - no action 1 _D Interrupt flag active, write 1 to clear the flag, read SPISF for details Reset: 0 _H
WK	1	rw1c	Wake interrupt flag Only set if device generates an interrupt when leaving SLEEP state. 0 _D No interrupt, write 0 - no action 1 _D Interrupt flag active, write 1 to clear the flag, read WKSF for details Reset: 0 _H

Field	Bits	Type	Description
SYS	0	rw1c	System interrupt flag 0 _D No interrupt, write 0 - no action 1 _D Interrupt flag active, write 1 to clear the flag, read SYSSF for details Reset: 0 _H

System status flags *R2)

SYSSF	Offset	Reset Value
System status flags *R2)	1D _H	00 _H

7	6	5	4	3	2	1	0
nu		NO_OP	TRFAIL	ERRMISS	FWDE	WWDE	CFGE
		rw1c	rw1c	rw1c	rw1c	rw1c	rw1c

Field	Bits	Type	Description
nu	7:6	none	Reset: 0 _H
NO_OP	5	rw1c	State transition request failure flag Requested state transition via DEVCTRL & DEVCTRLN could not be performed because of wrong protocol. 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
TRFAIL	4	rw1c	Transition to low power failed flag Transition to low power failed either due to the QUC current monitor, WAK high level or a rising edge on ENA during TRDEL time. 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
ERRMISS	3	rw1c	MCU error miss status flag Set only when SYSPCFG1.ERRRECEN='1' 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
FWDE	2	rw1c	Functional watchdog error interrupt flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H

Field	Bits	Type	Description
WWDE	1	rw1c	Window watchdog error interrupt flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
CFGE	0	rw1c	Protected configuration double bit error flag Double bit error occurred on protected configuration register. Status registers shall be read in order to determine which configuration has changed. 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H

Wakeup status flags *R2)

WKSF	Offset	Reset Value
Wakeup status flags *R2)	1E _H	00 _H

7	6	5	4	3	2	1	0
	nu		WKSPI	WKTIM	CMON	ENA	WAK
			rw1c	rw1c	rw1c	rw1c	rw1c

Field	Bits	Type	Description
nu	7:5	none	Reset: 0 _H
WKSPI	4	rw1c	Wakeup from SLEEP by SPI flag (GoToWAKE) 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
WKTIM	3	rw1c	Wake timer wakeup flag Bit will also be set if STANDBY state left because of wake timer expired. 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
CMON	2	rw1c	QUC current monitor threshold wakeup flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H

Field	Bits	Type	Description
ENA	1	rw1c	ENA signal wakeup flag Bit will also be set if FAILSAFE or STANDBY state left because of ENA. 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
WAK	0	rw1c	WAK signal wakeup flag Bit will also be set if FAILSAFE or STANDBY state left because of WAK. 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H

SPI status flags *R2)

SPISF	Offset	Reset Value
SPI status flags *R2)	1F_H	00_H

7	6	5	4	3	2	1	0
	nu		LOCK	DURE	ADDRE	LENE	PARE
			rw1c	rw1c	rw1c	rw1c	rw1c

Field	Bits	Type	Description
nu	7:5	none	Reset: 0 _H
LOCK	4	rw1c	LOCK or UNLOCK procedure error flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
DURE	3	rw1c	SPI frame duration error flag SCS low for more than 2 ms. 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
ADDRE	2	rw1c	SPI address invalid flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
LENE	1	rw1c	SPI frame length invalid flag Number of detected SPI clock cycles different than 16. 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H

Field	Bits	Type	Description
PARE	0	rw1c	SPI frame parity error flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H

Monitor status flags 0 *R1)**MONSF0****Offset****Reset Value****Monitor status flags 0 *R1)****20_H****00_H**

7	6	5	4	3	2	1	0
TRK2SG	TRK1SG	VREFSG	COMSG	VCORESG	STBYSG	UCSG	PREGSG
rw1c	rw1c	rw1c	rw1c	rw1c	rw1c	rw1c	rw1c

Field	Bits	Type	Description
TRK2SG	7	rw1c	Tracker2 short to ground status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
TRK1SG	6	rw1c	Tracker1 short to ground status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
VREFSG	5	rw1c	Voltage reference short to ground status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
COMSG	4	rw1c	Communication LDO short to ground status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
VCORESG	3	rw1c	Core voltage short to ground status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
STBYSG	2	rw1c	Standby LDO short to ground status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
UCSG	1	rw1c	uC LDO short to ground status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
PREGSG	0	rw1c	Pre-regulator voltage short to ground status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H

Monitor status flags 1 *R1)**MONSF1****Offset****Reset Value****Monitor status flags 1 *R1)****21_H****00_H**

7	6	5	4	3	2	1	0
TRK2OV	TRK1OV	VREFOV	COMOV	VCOREOV	STBYOV	UCOV	PREGOV
rw1c	rw1c	rw1c	rw1c	rw1c	rw1c	rw1c	rw1c

Field	Bits	Type	Description
TRK2OV	7	rw1c	Tracker2 over voltage status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
TRK1OV	6	rw1c	Tracker1 over voltage status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
VREFOV	5	rw1c	Voltage reference over voltage status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
COMOV	4	rw1c	Communication LDO over voltage status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
VCOREOV	3	rw1c	Core voltage over voltage status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
STBYOV	2	rw1c	Standby LDO over voltage status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
UCOV	1	rw1c	uC LDO over voltage status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
PREGOV	0	rw1c	Pre-regulator voltage over voltage status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H

Monitor status flags 2 *R2)**MONSF2****Offset****Reset Value****Monitor status flags 2 *R2)****22_H****00_H**

7	6	5	4	3	2	1	0
TRK2UV	TRK1UV	VREFUV	COMUV	VCOREUV	STBYUV	UCUV	PREGUV
rw1c	rw1c	rw1c	rw1c	rw1c	rw1c	rw1c	rw1c

Field	Bits	Type	Description
TRK2UV	7	rw1c	Tracker2 under voltage status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
TRK1UV	6	rw1c	Tracker1 under voltage status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
VREFUV	5	rw1c	Voltage reference under voltage status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
COMUV	4	rw1c	Communication LDO under voltage status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
VCOREUV	3	rw1c	Core voltage under voltage status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
STBYUV	2	rw1c	Standby LDO under voltage status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
UCUV	1	rw1c	uC LDO under voltage status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
PREGUV	0	rw1c	Pre-regulator voltage under voltage status flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H

Monitor status flags 3 *R1)**MONSF3****Offset****Reset Value****Monitor status flags 3 *R1)****23_H****00_H**

7	6	5	4	3	2	1	0
BIASHI	BIASLOW	BG12OV	BG12UV		nu		VBATOV
rw1c	rw1c	rw1c	rw1c				

Field	Bits	Type	Description
BIASHI	7	rw1c	Bias current too high flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
BIASLOW	6	rw1c	Bias current too low flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
BG12OV	5	rw1c	Bandgap comparator over voltage condition flag (VBG1 ≥ VBG2 + 4%) 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
BG12UV	4	rw1c	Bandgap comparator under voltage condition flag (VBG1 ≤ VBG2 - 4%) 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
nu	3:1	none	Reset: 0 _H
VBATOV	0	rw1c	Supply voltage VSx over voltage flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H

Over temperature failure status flags *R1)**OTFAIL****Offset****Reset Value****Over temperature failure status flags *R1)****24_H****00_H**

7	6	5	4	3	2	1	0
MON	nu		COM	nu		UC	PREG
rw1c			rw1c		rw1c		rw1c

Field	Bits	Type	Description
MON	7	rw1c	Monitoring over temperature flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
nu	6:5	none	Reset: 0 _H
COM	4	rw1c	Communication LDO over temperature flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
nu	3:2	none	Reset: 0 _H
UC	1	rw1c	uC LDO over temperature flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
PREG	0	rw1c	Pre-regulator over temperature flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H

Over temperature warning status flags *R2)

OTWRNSF

Offset

Reset Value

Over temperature warning status flags *R2)

25_H00_H

7	6	5	4	3	2	1	0
nu	VREF	COM	nu	STDBY	UC	PREG	
	rw1c	rw1c		rw1c	rw1c	rw1c	

Field	Bits	Type	Description
nu	7:6	none	Reset: 0 _H
VREF	5	rw1c	Voltage reference over load flag (over current for more than 1ms) 0 _D Write 0 no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
COM	4	rw1c	Communication LDO over temperature warning flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
nu	3	none	Reset: 0 _H
STDBY	2	rw1c	Standby LDO over load flag (over current for more than 1ms) 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
UC	1	rw1c	uC LDO over temperature warning flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H
PREG	0	rw1c	Pre-regulator over temperature warning flag 0 _D Write 0 - no action 1 _D Event detected, write 1 to clear the flag Reset: 0 _H

Voltage monitor status *R2)

VMONSTAT	Offset	Reset Value
Voltage monitor status *R2)	26 _H	00 _H

7	6	5	4	3	2	1	0
TRK2ST	TRK1ST	VREFST	COMST	VCOREST	STBYST	nu	
r	r	r	r	r	r		

Field	Bits	Type	Description
TRK2ST	7	r	Tracker2 voltage ready status 0 _D Voltage is out of range or not enabled 1 _D Voltage is OK Reset: 0 _H
TRK1ST	6	r	Tracker1 voltage ready status 0 _D Voltage is out of range or not enabled 1 _D Voltage is OK Reset: 0 _H
VREFST	5	r	Voltage reference voltage ready status 0 _D Voltage is out of range or not enabled 1 _D Voltage is OK Reset: 0 _H
COMST	4	r	Communication LDO voltage ready status 0 _D Voltage is out of range or not enabled 1 _D Voltage is OK Reset: 0 _H
VCOREST	3	r	Core voltage ready status 0 _D Voltage is out of range or not enabled 1 _D Voltage is OK Reset: 0 _H
STBYST	2	r	Standby LDO voltage ready status 0 _D Voltage is out of range or not enabled 1 _D Voltage is OK Reset: 0 _H
nu	1:0	none	Reset: 0 _H

Protection status *R1)

PROTSTAT

Offset

Reset Value

Protection status *R1)

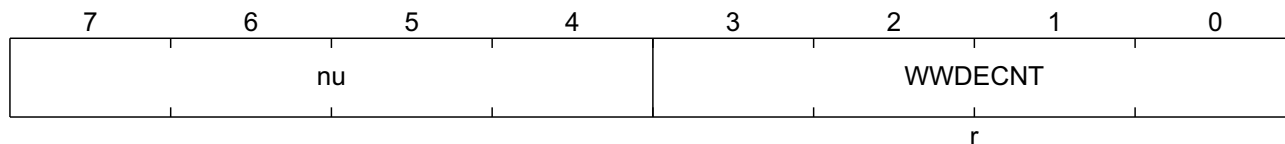
28_H01_H

7	6	5	4	3	2	1	0
KEY4OK	KEY3OK	KEY2OK	KEY1OK		nu		LOCK
r	r	r	r				r

Field	Bits	Type	Description
KEY4OK	7	r	Key4 ok status Information about validity of the 4th received protection key byte 0 _D Key not valid 1 _D Key valid Reset: 0 _H
KEY3OK	6	r	Key3 ok status Information about validity of the 3rd received protection key byte 0 _D Key not valid 1 _D Key valid Reset: 0 _H
KEY2OK	5	r	Key2 ok status Information about validity of the 2nd received protection key byte 0 _D Key not valid 1 _D Key valid Reset: 0 _H
KEY1OK	4	r	Key1 ok status Information about validity of the 1st received protection key byte 0 _D Key not valid 1 _D Key valid Reset: 0 _H
nu	3:1	none	Reset: 0 _H
LOCK	0	r	Protected register lock status 0 _D Access is unlocked 1 _D Access is locked Reset: 1 _H

Window watchdog status *R3)

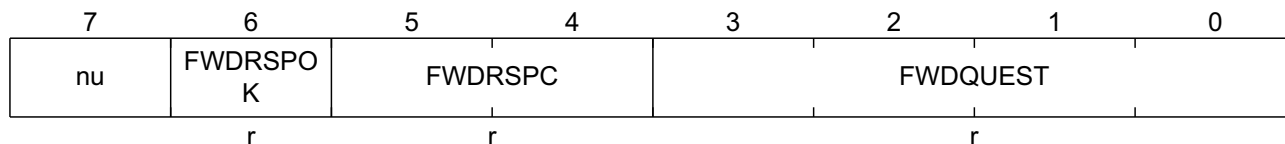
WWDSTAT **Offset**
Window watchdog status *R3) **29_H** **Reset Value**
00_H



Field	Bits	Type	Description
nu	7:4	none	Reset: 0 _H
WWDECNT	3:0	r	Window watchdog error counter status Reset: 0 _H

Functional watchdog status 0 *R3)

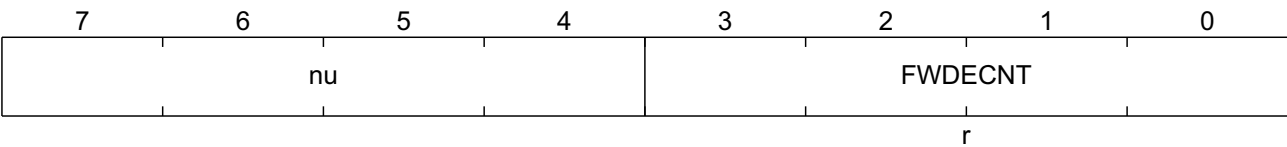
FWDSTAT0 **Offset**
Functional watchdog status 0 *R3) **2A_H** **Reset Value**
30_H



Field	Bits	Type	Description
nu	7	none	Reset: 0 _H
FWDRSPOK	6	r	Functional watchdog response check error status 0 _D Response message is wrong 1 _D All received bytes in response message are correct Reset: 0 _H
FWDSPC	5:4	r	Functional watchdog response counter value Reset: 3 _H
FWDQUEST	3:0	r	Functional watchdog question Reset: 0 _H

Functional watchdog status 1 *R3)

FWDSTAT1	Offset	Reset Value
Functional watchdog status 1 *R3)	2B _H	00 _H



Field	Bits	Type	Description
nu	7:4	none	Reset: 0 _H
FWDECNT	3:0	r	Functional watchdog error counter value Reset: 0 _H

ABIST control 0 *R2)

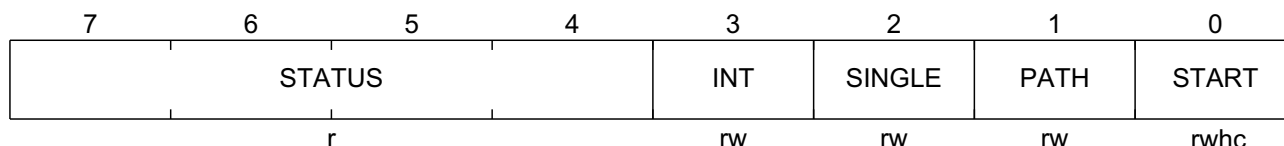
ABIST_CTRL0

ABIST control0 *R2)

Offset

2C_H

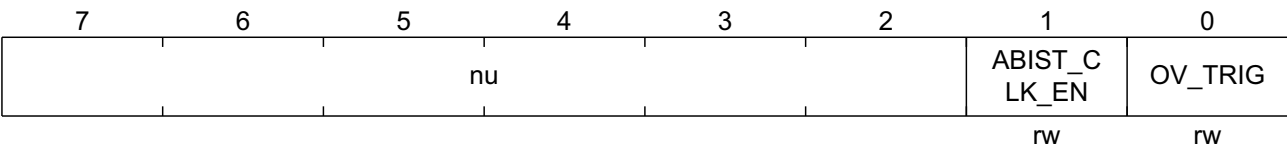
Reset Value

00_H

Field	Bits	Type	Description
STATUS	7:4	r	ABIST global error status ABIST status information after requested operation has been performed, information shall only be considered valid, once START bit is cleared. Bits have different reset class than whole register. The Bits are reset according to *R1). 5 _D Selected ABIST operation performed with no errors 10 _D Selected ABIST operation performed with errors, check respective SELECT registers Reset: 0 _H
INT	3	rw	Safety path selection Select whether safe state or interrupt related comparator shall be tested. 0 _D safe state related comparators shall be tested 1 _D interrupt related comparators shall be tested Reset: 0 _H
SINGLE	2	rw	ABIST Sequence selection Select whether a single comparator shall be tested or all comparators in predefined sequence 0 _D Predefined sequence 1 _D Single comparator test Reset: 0 _H
PATH	1	rw	Full path test selection Select the path which should be covered by ABIST operation 0 _D Comparator only 1 _D Comparator and corresponding deglitching logic, shall be selected in case contribution to respective safety measure needs to be tested Reset: 0 _H
START	0	rwhc	Start ABIST operation The ABIST operation itself will be started. This bit is cleared after ABIST operation has been performed 0 _D Operation done 1 _D Start operation Reset: 0 _H

ABIST control 1 *R2)

ABIST_CTRL1	Offset	Reset Value
ABIST control1 *R2)	2D _H	00 _H



Field	Bits	Type	Description
nu	7:2	none	Reset: 00 _H
ABIST_CLK_EN	1	rw	ABIST clock check enable Select ABIST clock to check its functionality 0 _D Disable 1 _D Enable Reset: 0 _H
OV_TRIG	0	rw	Overvoltage trigger for secondary internal monitor enable 0 _D Disable 1 _D Enable Reset: 0 _H

ABIST select 0 *R2)

ABIST_SELECT0
ABIST select 0 *R2)

Offset
2E_H

Reset Value
00_H

7	6	5	4	3	2	1	0
TRK2OV	TRK1OV	VREFOV	COMOV	VCOREOV	STBYOV	UCOV	PREGOV
rwhu	rwhu	rwhu	rwhu	rwhu	rwhu	rwhu	rwhu

Field	Bits	Type	Description
TRK2OV	7	rwhu	Select TRK2 OV comparator for ABIST operation 0 _D Not selected 1 _D Selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this comparator Reset: 0 _H
TRK1OV	6	rwhu	Select TRK1 OV comparator for ABIST operation 0 _D Not selected 1 _D Selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this comparator Reset: 0 _H
VREFOV	5	rwhu	Select VREF OV comparator for ABIST operation 0 _D Not selected 1 _D Selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this comparator Reset: 0 _H
COMOV	4	rwhu	Select COM OV comparator for ABIST operation 0 _D Not selected 1 _D Selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this comparator Reset: 0 _H
VCOREOV	3	rwhu	Select Core voltage OV comparator for ABIST operation 0 _D Not selected 1 _D Selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this comparator Reset: 0 _H
STBYOV	2	rwhu	Select Standby LDO OV comparator for ABIST operation 0 _D Not selected 1 _D Selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this comparator Reset: 0 _H
UCOV	1	rwhu	Select uC LDO OV comparator for ABIST operation 0 _D Not selected 1 _D Selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this comparator Reset: 0 _H

Field	Bits	Type	Description
PREGOV	0	rwhu	Select Pre-regulator OV comparator for ABIST operation 0 _D Not selected 1 _D Selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this comparator Reset: 0 _H

ABIST select 1 *R2)

ABIST_SELECT1	Offset	Reset Value
ABIST select 1 *R2)	2F_H	00_H

7	6	5	4	3	2	1	0
TRK2UV	TRK1UV	VREFUV	COMUV	VCOREUV	STBYUV	UCUV	PREGUV
rwhu	rwhu	rwhu	rwhu	rwhu	rwhu	rwhu	rwhu

Field	Bits	Type	Description
TRK2UV	7	rwhu	Select TRK2 UV comparator for ABIST operation 0 _D Not selected 1 _D Selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this comparator Reset: 0 _H
TRK1UV	6	rwhu	Select TRK1 UV comparator for ABIST operation 0 _D Not selected 1 _D Selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this voltage Reset: 0 _H
VREFUV	5	rwhu	Select VREF UV comparator for ABIST operation 0 _D Not selected 1 _D Selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this comparator Reset: 0 _H
COMUV	4	rwhu	Select COM UV comparator for ABIST operation 0 _D Not selected 1 _D Selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this comparator Reset: 0 _H
VCOREUV	3	rwhu	Select VCORE UV comparator for ABIST operation 0 _D Not selected 1 _D selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this comparator Reset: 0 _H

Field	Bits	Type	Description
STBYUV	2	rwhu	Select STBY UV comparator for ABIST operation 0_D Not selected 1_D selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this comparator Reset: 0_H
UCUV	1	rwhu	Select uC UV comparator for ABIST operation 0_D Not selected 1_D selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this comparator Reset: 0_H
PREGUV	0	rwhu	Select pre regulator UV comparator for ABIST operation 0_D Not selected 1_D Selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this comparator Reset: 0_H

ABIST select 2 *R2)

ABIST_SELECT2	Offset	Reset Value
ABIST select 2 *R2)	30_H	00_H

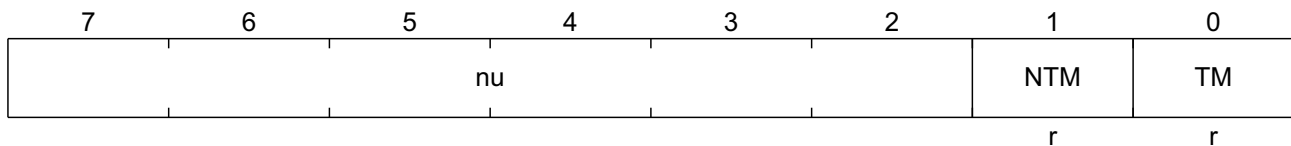
7	6	5	4	3	2	1	0
BIASHI	BIASLOW	BG12OV	BG12UV	INTOV	nu		VBATOV
rwhu	rwhu	rwhu	rwhu	rwhu			rwhu

Field	Bits	Type	Description
BIASHI	7	rwhu	Select bias current too high 0_D Not selected 1_D Selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this comparator Reset: 0_H
BIASLOW	6	rwhu	Select bias current too low 0_D Not selected 1_D Selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this comparator Reset: 0_H
BG12OV	5	rwhu	Select bandgap comparator OV condition $(VBG1 \geq VBG2 + 4\%)$ 0_D Not selected 1_D Selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this comparator Reset: 0_H

Field	Bits	Type	Description
BG12UV	4	rwhu	Select bandgap comparator UV condition ($VBG1 \leq VBG2 - 4\%$) 0 _D Not selected 1 _D Selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this comparator Reset: 0 _H
INTOV	3	rwhu	Select internal supply OV condition 0 _D Not selected 1 _D Selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this comparator Reset: 0 _H
nu	2:1	none	Reset: 0 _H
VBATOV	0	rwhu	Select supply VSx overvoltage 0 _D Not selected 1 _D Selected, bit will be cleared upon successful ABIST operation on comparator, bit will be set in case of ABIST fail for this comparator Reset: 0 _H

Global testmode *R2)

GTM	Offset	Reset Value
Global testmode *R2)	3F_H	02_H

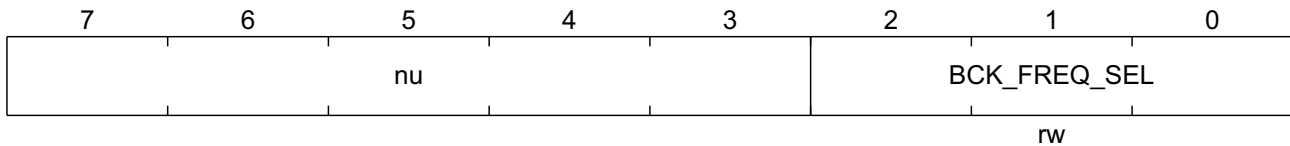


Field	Bits	Type	Description
nu	7:2	none	Reset: 00 _H
NTM	1	r	Test mode inverted status 0 _D Device is in test mode 1 _D Device is in normal mode Reset: 1 _H
TM	0	r	Test mode status 0 _D Device is in normal mode 1 _D Device is in test mode Reset: 0 _H

13.4.2 Buck registers

Buck switching frequency change *R2)

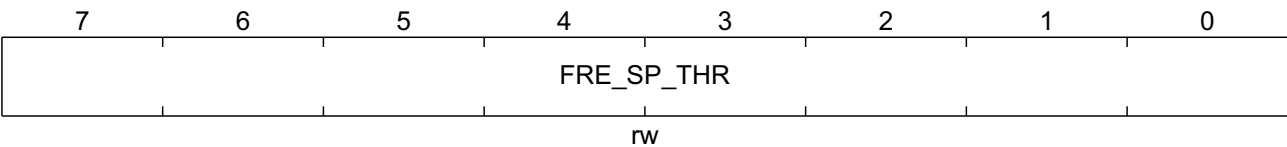
BCK_FREQ_CHANGE	Offset	Reset Value
Buck switching frequency change *R2)	31 _H	00 _H



Field	Bits	Type	Description
nu	7:3	none	Reset: 00 _H
BCK_FREQ_SEL	2:0	rw	BUCK switching frequency change For hi and low switching mode. New value needs to be validated via data_valid procedure 7 _D Change buck frequency by approx. -4.5% from $f_{OSC,step-down}$ 6 _D Change buck frequency by approx. -3.0% from $f_{OSC,step-down}$ 5 _D Change buck frequency by approx. -1.5% from $f_{OSC,step-down}$ 4 _D No change 3 _D Change buck frequency by approx. +4.5% from $f_{OSC,step-down}$ 2 _D Change buck frequency by approx. +3.0% from $f_{OSC,step-down}$ 1 _D Change buck frequency by approx. +1.5% from $f_{OSC,step-down}$ 0 _D No Change Reset: 0 _H

Buck Frequency spread *R2)

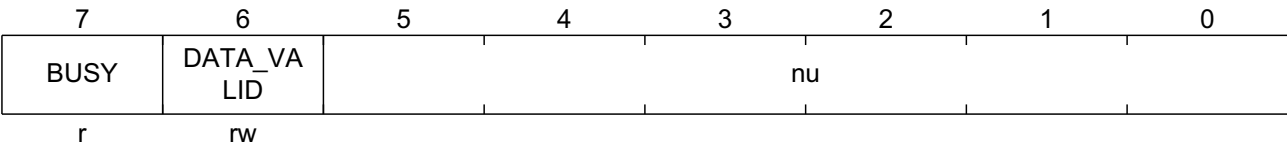
BCK_FRE_SPREAD	Offset	Reset Value
Buck Frequency spread *R2)	32 _H	00 _H



Field	Bits	Type	Description
FRE_SP_THR	7:0	rw	Spread spectrum Select the percentage of frequency spread(+/-). The mean frequency is reduced by the percentage as well keeping the maximum frequency at the nominal frequency selected by FRE. 00 _H No spread 2B _H 1% 55 _H 2% 80 _H 3% AA _H 4% D5 _H 5% FF _H 6% Reset: 00 _H

Buck main control *R2)

BCK_MAIN_CTRL	Offset	Reset Value
Buck main control *R2)	33 _H	00 _H



Field	Bits	Type	Description
BUSY	7	r	DATA_VALID parameter update ready status 0 _D update done 1 _D update ongoing Reset: 0 _H

Field	Bits	Type	Description
DATA_VALID	6	rw	Enable buck update Update Command to load new parameter for stepdown regulator (after configuration write 1 to update and write 0 after BUSY flag is cleared to proceed operation) 0 _D No action 1 _D Load new parameters Reset: 0 _H
nu	5:0	none	Reset: 00 _H

13.5 Electrical Characteristics

Table 23 Electrical Characteristics: SPI signals

$V_{VS} = 6.0\text{ V to }40\text{ V}$, $T_j = -40^\circ\text{C to }+150^\circ\text{C}$, all voltages with respect to ground, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note / Test Condition	Number
		Min.	Typ.	Max.			
Pin SCS, Chip Select							
Chip select valid high level	$V_{SCS, hi}$	3.6	—	—	V	V_{SCS} increasing, $V_{QUC} = 5.0\text{ V}$	P_13.5.1
Chip select valid low level	$V_{SCS, lo}$	—	—	0.8	V	V_{SCS} decreasing, $V_{QUC} = 5.0\text{ V}$	P_13.5.2
Chip select hysteresis	$V_{SCS, hyst}$	—	350	—	mV	$V_{QUC} = 5.0\text{ V}$	P_13.5.3
Chip select valid high level	$V_{SCS, hi}$	2.0	—	—	V	V_{SCS} increasing, $V_{QUC} = 3.3\text{ V}$	P_13.5.4
Chip select valid low level	$V_{SCS, lo}$	—	—	0.8	V	V_{SCS} decreasing, $V_{QUC} = 3.3\text{ V}$	P_13.5.5
Chip select hysteresis	$V_{SCS, hyst}$	—	160	—	mV	$V_{QUC} = 3.3\text{ V}$	P_13.5.6
Chip select pull-up current	I_{SCS}	-175	-120	—	μA	$V_{SCS} = 0\text{ V}$	P_13.5.7
Chip select input capacitance	C_{SCS}	—	4	15	pF	1)	P_13.5.8
Pin SCL, Clock							
Clock signal valid high level	$V_{SCL, hi}$	3.6	—	—	V	V_{SCL} increasing, $V_{QUC} = 5.0\text{ V}$	P_13.5.9
Clock signal valid low level	$V_{SCL, lo}$	—	—	0.8	V	V_{SCL} decreasing, $V_{QUC} = 5.0\text{ V}$	P_13.5.10
Clock hysteresis	$V_{SCL, hyst}$	—	350	—	mV	$V_{QUC} = 5.0\text{ V}$	P_13.5.11
Clock signal valid high level	$V_{SCL, hi}$	2.0	—	—	V	V_{SCL} increasing, $V_{QUC} = 3.3\text{ V}$	P_13.5.12
Clock signal valid low level	$V_{SCL, lo}$	—	—	0.8	V	V_{SCL} decreasing, $V_{QUC} = 3.3\text{ V}$	P_13.5.13
Clock hysteresis	$V_{SCL, hyst}$	—	160	—	mV	$V_{QUC} = 3.3\text{ V}$	P_13.5.14
Clock signal pull-down current	I_{SCL}	—	150	330	μA	$V_{SCL} = V_{QUC}$	P_13.5.15
Clock input capacitance	C_{SCL}	—	4	15	pF	1)	P_13.5.16
Pin SDI, Data Input, MOSI							
Data input valid high level	$V_{SDI, hi}$	3.6	—	—	V	V_{SDI} increasing, $V_{QUC} = 5.0\text{ V}$	P_13.5.17
Data input valid low level	$V_{SDI, lo}$	—	—	0.8	V	V_{SDI} decreasing, $V_{QUC} = 5.0\text{ V}$	P_13.5.18
Data input hysteresis	$V_{SDI, hyst}$	—	350	—	mV	$V_{QUC} = 5.0\text{ V}$	P_13.5.19
Data input valid high level	$V_{SDI, hi}$	2.0	—	—	V	V_{SDI} increasing, $V_{QUC} = 3.3\text{ V}$	P_13.5.20

Table 23 Electrical Characteristics: SPI signals (cont'd)

$V_{VS} = 6.0 \text{ V}$ to 40 V , $T_j = -40^\circ\text{C}$ to $+150^\circ\text{C}$, all voltages with respect to ground, positive current flowing into pin (unless otherwise specified)

Parameter	Symbol	Values			Unit	Note / Test Condition	Number
		Min.	Typ.	Max.			
Data input valid low level	$V_{SDI, lo}$	—	—	0.8	V	V_{SDI} decreasing, $V_{QUC} = 3.3 \text{ V}$	P_13.5.21
Data input hysteresis	$V_{SDI, hyst}$	—	160	—	mV	$V_{QUC} = 3.3 \text{ V}$	P_13.5.22
Data input signal pull-down current	I_{SDI}	—	150	330	μA	$V_{SDI} = V_{QUC}$	P_13.5.23
Data input capacitance	C_{SDI}	—	4	15	pF	¹⁾	P_13.5.24

Pin SDO, Data Output, MISO

Data output high level TLF35584xxVS1	$V_{SDO, hi}$	4.0	—	—	V	$V_{QUC} = 5.0 \text{ V}$, $I_{SDO} = -9 \text{ mA}$	P_13.5.25
Data output low level TLF35584xxVS1	$V_{SDO, lo}$	—	—	0.7	V	$V_{QUC} = 5.0 \text{ V}$ $I_{SDO} = 7 \text{ mA}$	P_13.5.26
Data output high level TLF35584xxVS2	$V_{SDO, hi}$	2.3	—	—	V	$V_{QUC} = 3.3 \text{ V}$ $I_{SDO} = -7 \text{ mA}$	P_13.5.27
Data output low level TLF35584xxVS2	$V_{SDO, lo}$	—	—	0.7	V	$V_{QUC} = 3.3 \text{ V}$ $I_{SDO} = 5.5 \text{ mA}$	P_13.5.28
Data output rise time ¹⁾	$t_{SDO, rise}$	—	—	25	ns	$C_{SDO, load} = 50 \text{ pF}$	P_13.5.29
Data output fall time ¹⁾	$t_{SDO, fall}$	—	—	25	ns	$C_{SDO, load} = 50 \text{ pF}$	P_13.5.30
Data output tristate capacitance	$C_{SDO, tri}$	—	4	15	pF	¹⁾	P_13.5.31
Data output tristate leakage	$I_{SDO, tr, leaki}$	-10	—	10	μA	—	P_13.5.32

¹⁾ Specified by design, not subject to production test